

SOFTWARE ENGINEERING LAB 3

Multi-Vendor Artisan E-Commerce Marketplace

Agile Project Planning and Sprint Simulation using Jira

Student	Rachit S Mane
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Problem Statement	Multi-Vendor Artisan E-Commerce Marketplace
Jira Project	Multi-Vendor Artisan E-Commerce Marketplace (key: MVAECM)
Jira Board	MVAECM board
Epics	4 (MVAECM-1 to MVAECM-4)
User Stories	12 (MVAECM-5 to MVAECM-16)
Sprints	MVAECM Sprint 1, MVAECM Sprint 2

All Epics, User Stories, story points, priorities, sprint activity and reports shown in this document are taken from the student's own Jira project.

A. Epic and User Story Table

The table below lists all 4 Epics created in the Jira project and every User Story belonging to each Epic, together with its Jira issue key, story points, priority and the sprint in which it was worked on. All values are taken directly from the Jira evidence reproduced later in this document.

Epic	Epic Key	User Story	Issue Key	Story Points	Priority	Sprint
Vendor Storefront & Product Catalog	MVAECM-1	Create artisan storefront	MVAECM-5	3	High	Sprint 2
		Manage product catalog	MVAECM-6	5	High	Sprint 2
		Display product images	MVAECM-7	3	Medium	Backlog
Product Discovery & Shopping	MVAECM-2	Browse artisan products	MVAECM-8	3	High	Sprint 2
		View product details	MVAECM-9	3	High	Sprint 2
		Add products to cart	MVAECM-10	5	High	Sprint 1
Multi-Vendor Checkout & Payments	MVAECM-3	Place customer order	MVAECM-11	5	Highest	Sprint 1
		Process checkout payment	MVAECM-12	5	Highest	Sprint 1
		Track order status	MVAECM-13	3	Medium	Backlog
Order & Vendor Management	MVAECM-4	Split payment among vendors	MVAECM-14	8	Highest	Sprint 1
		Deduct platform commission	MVAECM-15	5	High	Backlog
		View vendor payout	MVAECM-16	3	Medium	Backlog
Total		12 User Stories across 4 Epics		51		

Source of the values above: Epic list - Figure 1; user stories, story points and priorities - Figures 2 to 4; sprint membership - Figures 6, 10, 11 and 14.

B. Evaluation Criteria and Where the Evidence Appears

Evaluation Criterion	Marks	Evidence in this document
Create Stories in Jira	4	Epic and User Story table (Part A); Section 1, Figures 1 and 2
Assign Story Points Planning Poker introduction	3	Section 2, Figure 3; Story Points column of the table in Part A
Prioritize & Rank Backlog Set priority for each story; reorder backlog	2	Section 3, Figure 4 (priorities set and backlog reordered by rank)
Run a Mock Sprint Create Sprint 1; simulate progress; add comments	3	Section 4, Figures 5 and 6 (sprint created and started); Section 5, Figures 7 to 14 (progress simulation and comments); Section 6, Figures 15 to 18 (sprint summary)
Generate Burndown Chart	1	Section 7, Figures 19 and 20
Group Reflection	2	Section 8

1. Epics and User Stories

Four Epics (MVAECM-1 to MVAECM-4) were created for the Multi-Vendor Artisan E-Commerce Marketplace project, and twelve User Stories (MVAECM-5 to MVAECM-16) were created and linked to them, three stories per Epic.

The screenshot displays the Jira interface for the 'Multi-Vendor Artisan E-Commerce Marketplace' project. The 'MVAECM board' is shown in 'List' view. The board contains four epic items, each with a link icon and a title. All items are assigned to 'RACHIT MANE' and have a 'Medium' priority and 'To Do' status.

	Work	Assignee	Reporter	Priority	Status
> MVAECM-1	Vendor Storefront & Product Catalog	Unassigned	RACHIT MANE	Medium	To Do
> MVAECM-2	Product Discovery & Shopping	Unassigned	RACHIT MANE	Medium	To Do
> MVAECM-3	Multi-Vendor Checkout & Payments	Unassigned	RACHIT MANE	Medium	To Do
> MVAECM-4	Order & Vendor Management	Unassigned	RACHIT MANE	Medium	To Do

A notification in the bottom left corner states: 'Created Story "MVAECM-16"'. Below it, a link says 'View Story · Copy link'.

Figure 1 - The four Epics created in the Jira project: MVAECM-1 Vendor Storefront & Product Catalog, MVAECM-2 Product Discovery & Shopping, MVAECM-3 Multi-Vendor Checkout & Payments and MVAECM-4 Order & Vendor Management.

The screenshot displays the Jira interface for the 'Multi-Vendor Artisan E-Commerce Marketplace' project. The main backlog view shows twelve user stories, each with a checkbox, ID, title, status, priority, and assignee. The stories are color-coded by epic: blue for MVAECM-5 to MVAECM-10, purple for MVAECM-11 to MVAECM-13, and orange for MVAECM-14 to MVAECM-16. The right-hand panel provides a detailed view of the selected story, MVAECM-7 'Display product images', including its description, acceptance criteria, and attachments.

ID	Title	Status	Priority	Assignee
MVAECM-5	Create artisan storefront	To Do	3	
MVAECM-6	Manage product catalog	To Do	5	
MVAECM-7	Display product images	To Do	3	
MVAECM-8	Browse artisan products	To Do	3	
MVAECM-9	View product details	To Do	3	
MVAECM-10	Add products to cart	To Do	5	
MVAECM-11	Place customer order	To Do	5	
MVAECM-12	Process checkout payment	To Do	5	
MVAECM-13	Track order status	To Do	3	
MVAECM-14	Split payment among vendors	To Do	8	
MVAECM-15	Deduct platform commission	To Do	5	
MVAECM-16	View vendor payout	To Do	3	

Detail Panel for MVAECM-7: Display product images

Description: As an artisan vendor, I want to upload high-resolution product images so that shoppers can clearly view my products.

Acceptance Criteria:

- The vendor can upload product images.
- Images are displayed with the corresponding product.
- Shoppers can view the product images.

Attachments: Add attachment

Figure 2 - All twelve User Stories (MVAECM-5 to MVAECM-16) created in the backlog, each carrying its Epic colour tag. The detail panel shows story MVAECM-7 "Display product images" under Epic MVAECM-1, with its user-story description and acceptance criteria.

2. Story Points and Planning Poker Estimation

Planning Poker.

Story points for this project were estimated using the Planning Poker technique. In Planning Poker every team member privately selects an estimate for a user story from a Fibonacci-style deck (1, 2, 3, 5, 8, 13, ...); all estimates are revealed at the same time so that nobody anchors on anyone else's number, the highest and lowest estimates are discussed, and the round is repeated until the team converges on a single agreed value. The agreed value is then recorded against the story in Jira.

The values agreed for this project are 3, 5 and 8 - all Fibonacci values from the Planning Poker deck - and every one of the twelve user stories carries an estimate, giving a total of 51 story points for the product backlog. The estimates are visible in the Story Points column of the Jira backlog below, and are listed per story in the table in Part A.

The screenshot displays a Jira backlog for the 'MVAECM board'. The backlog contains 16 user stories, each with a checkbox, a title, a status (all are 'To Do'), and a story point value. The story points are: MVAECM-14 (8), MVAECM-11 (5), MVAECM-12 (5), MVAECM-10 (5), MVAECM-6 (5), MVAECM-5 (3), MVAECM-8 (3), MVAECM-9 (3), MVAECM-15 (5), MVAECM-7 (3), MVAECM-13 (3), and MVAECM-16 (3). The right sidebar shows the details for 'MVAECM-12: Process checkout payment', including a description and acceptance criteria.

Issue ID	Story Title	Story Points
MVAECM-14	Split payment among vendors	8
MVAECM-11	Place customer order	5
MVAECM-12	Process checkout payment	5
MVAECM-10	Add products to cart	5
MVAECM-6	Manage product catalog	5
MVAECM-5	Create artisan storefront	3
MVAECM-8	Browse artisan products	3
MVAECM-9	View product details	3
MVAECM-15	Deduct platform commission	5
MVAECM-7	Display product images	3
MVAECM-13	Track order status	3
MVAECM-16	View vendor payout	3

Process checkout payment (MVAECM-12) Details:

Description: As a shopper, I want to make a payment during checkout so that I can complete my purchase.

Acceptance Criteria:

- The shopper can initiate payment at checkout.
- The payment is associated with the order.
- The system confirms successful payment.
- An unsuccessful payment does not create an incorrectly completed order.

Figure 3 - Story points assigned to every user story in the Jira backlog (Story Points column): 8 points for MVAECM-14, 5 points for MVAECM-6, 10, 11, 12 and 15, and 3 points for MVAECM-5, 7, 8, 9, 13 and 16.

3. Priority and Backlog Ranking

A priority was set for every user story in Jira (Highest, High or Medium), and the backlog was then reordered so that the highest-priority and highest-value stories sit at the top and are pulled into the sprint first.

Backlog order before ranking (Figure 2): MVAECM-5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16 - the order in which the stories were created.

Backlog order after ranking (Figure 4): MVAECM-14, 11, 12, 10, 6, 5, 8, 9, 15, 7, 13, 16 - the three Highest-priority stories (MVAECM-14, 11, 12) have been moved to the top of the backlog, followed by the High-priority stories, with the Medium-priority stories ranked last.

The screenshot shows the Jira backlog for the 'Multi-Vendor Artisan E-Commerce Marketplace' project. The backlog is sorted by priority and value. The top three items are MVAECM-14, MVAECM-11, and MVAECM-12, all with a priority of 'Highest' (indicated by a double chevron icon). The bottom three items are MVAECM-5, MVAECM-8, and MVAECM-9, all with a priority of 'Medium' (indicated by an equals sign icon). The right sidebar shows the details for the selected item, MVAECM-12, titled 'Process checkout payment'.

Item	Priority	Value
MVAECM-14	Highest	8
MVAECM-11	Highest	5
MVAECM-12	Highest	5
MVAECM-10	High	5
MVAECM-6	High	5
MVAECM-5	Medium	3
MVAECM-8	Medium	3
MVAECM-9	Medium	3
MVAECM-15	High	5
MVAECM-7	High	3
MVAECM-13	High	3
MVAECM-16	High	3

Figure 4 - The reordered and ranked product backlog. The priority icon on each row shows the priority set for that story in Jira (double chevron = Highest, single chevron = High, equals sign = Medium).

4. Sprint 1

MVAECM Sprint 1 was created from the top of the ranked backlog and started with a duration of one week. Four user stories worth 23 story points were committed to the sprint: MVAECM-14 (8), MVAECM-11 (5), MVAECM-12 (5) and MVAECM-10 (5).

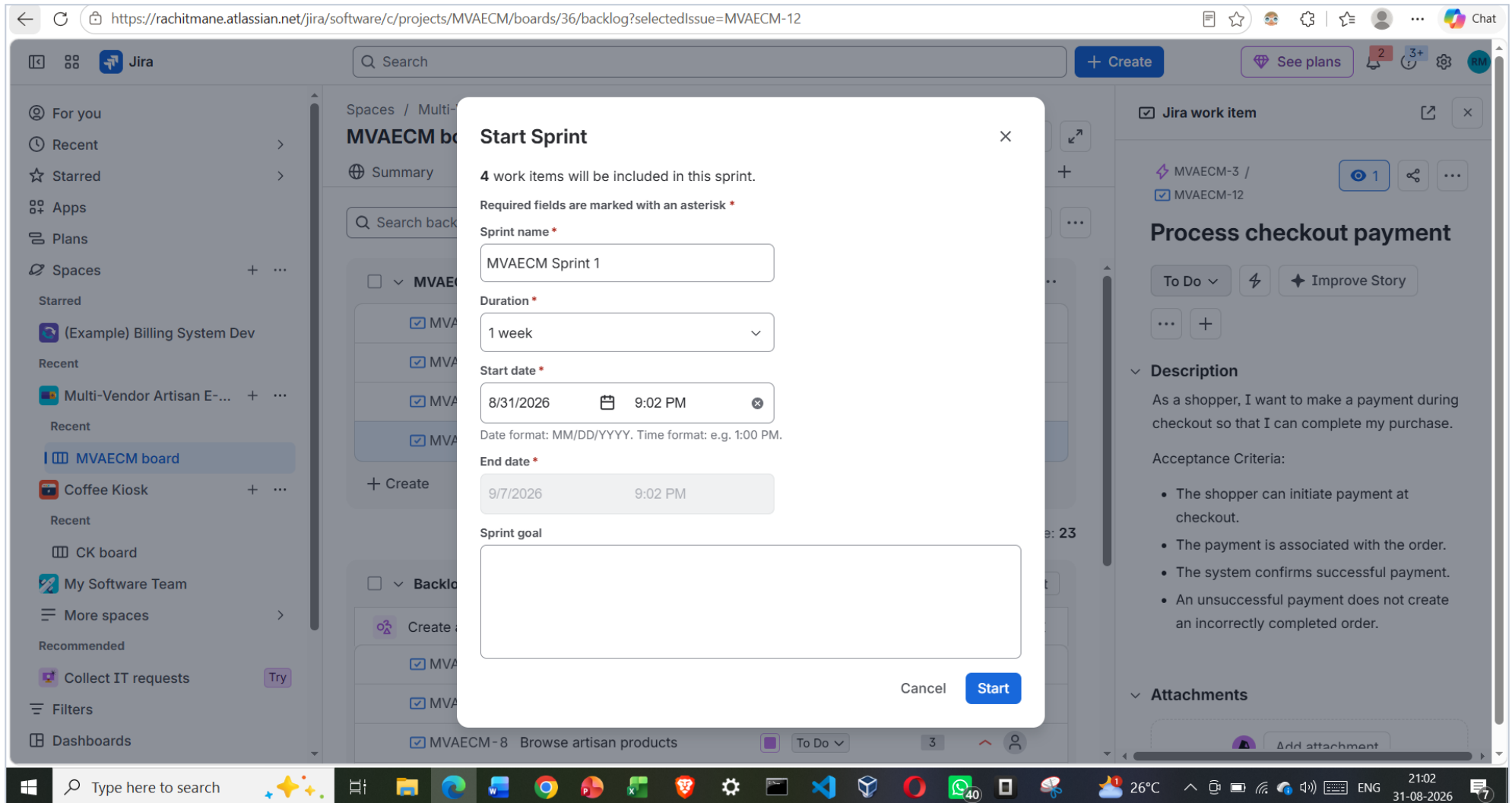


Figure 5 - Creating and starting MVAECM Sprint 1: sprint name "MVAECM Sprint 1", duration 1 week, start date 31/08/2026 9:02 PM, end date 07/09/2026 9:02 PM, with 4 work items included in the sprint.

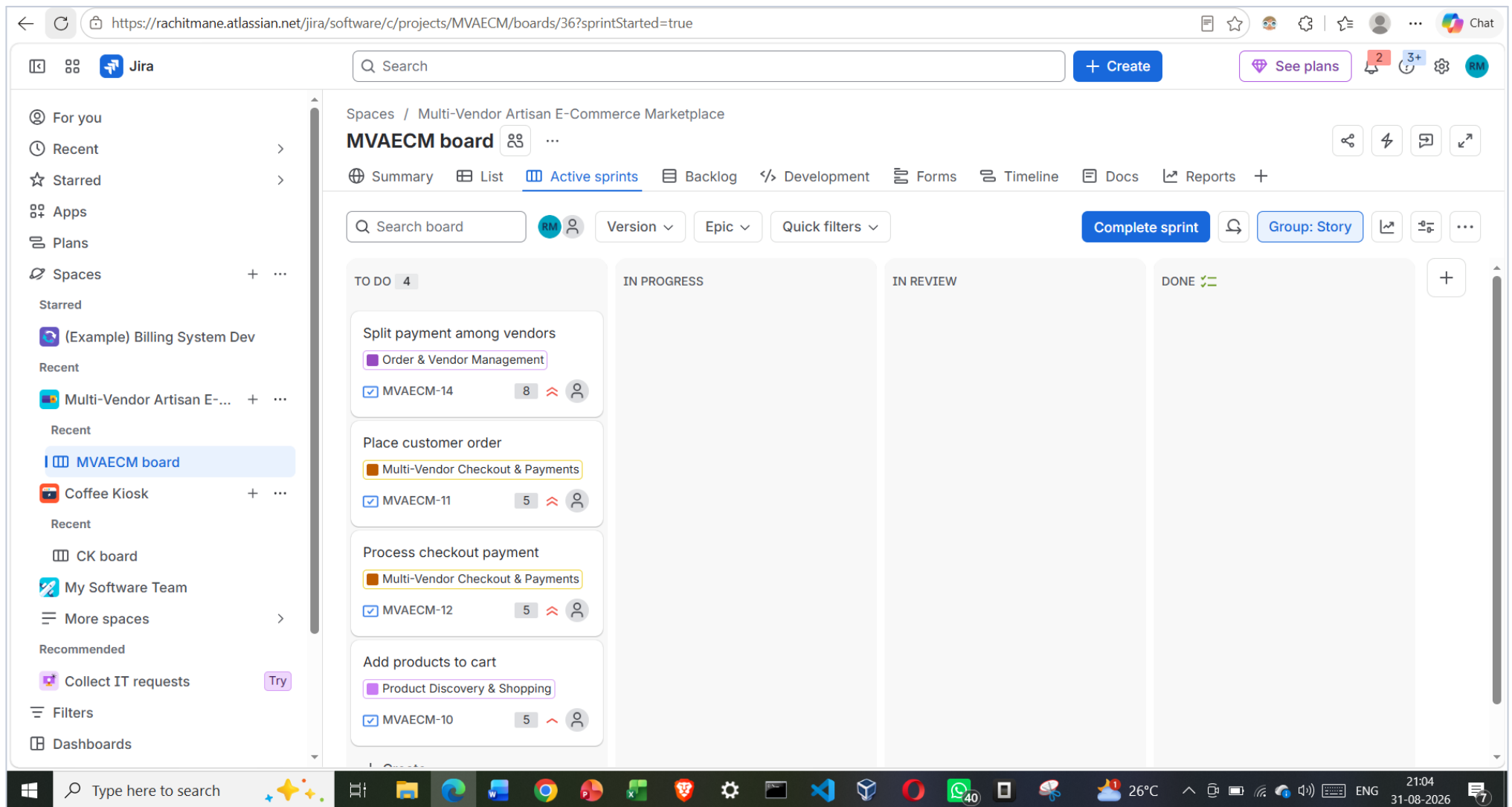


Figure 6 - MVAECM Sprint 1 started. The active sprint board shows the four committed stories in the TO DO column: MVAECM-14 Split payment among vendors (8), MVAECM-11 Place customer order (5), MVAECM-12 Process checkout payment (5) and MVAECM-10 Add products to cart (5).

5. Sprint Progress and Comments

Progress was simulated by moving the sprint stories across the board columns (To Do -> In Progress -> In Review -> Done) and by adding comments to the work items as the work progressed.

5.1 Sprint 1 progress and comments

The screenshot displays a Jira board for the 'Multi-Vendor Artisan E-Commerce Marketplace'. The board is organized into four columns: 'TO DO', 'IN PROGRESS', 'IN REVIEW', and 'DONE'. In the 'IN PROGRESS' column, two issues are visible: MVAECM-11 and MVAECM-12. MVAECM-11 is titled 'Place customer order' and MVAECM-12 is titled 'Process checkout payment'. Both issues have a 'Multi-Vendor Check...' label. A comment by RACHIT MANE is visible on the right side of the board, stating: 'Started implementation of the customer ordering workflow and validating the order creation process.' The left sidebar shows the Jira navigation menu with options like 'For you', 'Recent', 'Starred', 'Apps', 'Plans', 'Spaces', and 'Dashboards'. The top bar includes a search bar, a '+ Create' button, and a 'See plans' button. The bottom of the screen shows a Windows taskbar with various application icons and system information like temperature and time.

Figure 7 - Sprint 1 in progress: MVAECM-11 and MVAECM-12 moved to IN PROGRESS. Comment added by RACHIT MANE on MVAECM-11: "Started implementation of the customer ordering workflow and validating the order creation process."

← ↻ 🔒 https://rachitmane.atlassian.net/jira/software/c/projects/MVAECM/boards/36?selectedIssue=MVAECM-12&sprintStarted=true

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Spaces / Multi-Vendor Artisan E-Commerce Marketplace

MVAECM board Summary List Active sprints Backlog Development Forms More 3 +

Search board RM Version More Complete sprint Group: Story

TO DO 2 IN PROGRESS 2 IN REVIEW DONE

Split payment among vendors
Order & Vendor Ma...
8 MVAECM-14

Add products to cart
Product Discovery ...
5 MVAECM-10

Place customer order
Multi-Vendor Chec...
5 MVAECM-11

Process checkout payment
Multi-Vendor Chec...
5 MVAECM-12

+ Create

Jira work item

More fields Original estimate, Time tracki...

Automation Rule executions

Created 19 minutes ago Updated 1 minute ago Configure

Activity

Comments More

Add a comment...
Suggest a reply... Who is w >

Pro tip: press M to comment

RACHIT MANE
1 second ago

Working on checkout payment processing and validating successful and unsuccessful payment scenarios.

← 👍 😊 📎 ...

Type here to search 26°C 21:07 31-08-2026

Figure 8 - Comment added by RACHIT MANE on MVAECM-12: "Working on checkout payment processing and validating successful and unsuccessful payment scenarios."

← ↻ 🔒 https://rachitmane.atlassian.net/jira/software/c/projects/MVAECM/boards/36?sprintStarted=true

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Summary List **Active sprints** Backlog Development Forms Timeline Docs Reports +

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TO DO 2

Split payment among vendors

Order & Vendor Management

MVAECM-14 8

Add products to cart

Product Discovery & Shopping

MVAECM-10 5

+ Create

IN PROGRESS

IN REVIEW 1

Place customer order

Multi-Vendor Checkout & Payments

MVAECM-11 5

DONE 1 ✓

Process checkout payment

Multi-Vendor Checkout & Payments

MVAECM-12 5

+ 21:08 31-08-2026

Figure 9 - Sprint 1 progress: MVAECM-11 Place customer order moved to IN REVIEW and MVAECM-12 Process checkout payment moved to DONE.

← ↻ https://rachitmane.atlassian.net/jira/software/c/projects/MVAECM/boards/36?sprintStarted=true

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MVAECM board ...

Summary List **Active sprints** Backlog Development Forms Timeline Docs Reports +

Search board RM Version Epic Quick filters Complete sprint Group: Story

TO DO IN PROGRESS IN REVIEW DONE 4 ✓

+ Create

Process checkout payment
Multi-Vendor Checkout & Payments
MVAECM-12 5

Place customer order
Multi-Vendor Checkout & Payments
MVAECM-11 5

Split payment among vendors
Order & Vendor Management
MVAECM-14 8

Add products to cart
Product Discovery & Shopping
MVAECM-10 5

Windows Taskbar: Type here to search, 26°C, 21:09, 31-08-2026

Figure 10 - End of Sprint 1: all four committed stories (MVAECM-12, MVAECM-11, MVAECM-14 and MVAECM-10) completed in the DONE column.

5.2 Sprint 2 progress and comments

After Sprint 1 was completed, the next four stories from the ranked backlog were taken into MVAECM Sprint 2 (MVAECM-6, MVAECM-5, MVAECM-8 and MVAECM-9, worth 14 story points) and the same progress simulation and commenting was carried out.

The screenshot displays the Jira interface for the 'Multi-Vendor Artisan E-Commerce Marketplace' project. The 'MVAECM board' is active, showing a Kanban view with four columns: 'TO DO', 'IN PROGRESS', 'IN REVIEW', and 'DONE'. The 'TO DO' column contains four stories, each with a checkbox, a title, a category, and a story point value. The other columns are currently empty. The left sidebar shows the project's structure, including 'For you', 'Recent', 'Starred', 'Apps', 'Plans', 'Spaces', and 'Recommended'. The top bar includes a search bar, a '+ Create' button, and a 'See plans' link. The bottom of the screen shows a Windows taskbar with various application icons and system information.

Column	Story	Category	Points
TO DO	Manage product catalog	Vendor Storefront & Product Catalog	5
	Create artisan storefront	Vendor Storefront & Product Catalog	3
	Browse artisan products	Product Discovery & Shopping	3
	View product details	Product Discovery & Shopping	3
IN PROGRESS			
IN REVIEW			
DONE			

Figure 11 - MVAECM Sprint 2 board at the start of the sprint, with MVAECM-6 Manage product catalog (5), MVAECM-5 Create artisan storefront (3), MVAECM-8 Browse artisan products (3) and MVAECM-9 View product details (3) in the TO DO column.

The screenshot shows a Jira board for the 'Multi-Vendor Artisan E-Commerce Marketplace' project. The board is titled 'MVAECM board' and is in the 'Active sprints' view. The board has four columns: 'TO DO' (3 items), 'IN PROGRESS' (1 item), 'IN REVIEW' (0 items), and 'DONE' (0 items). The 'IN PROGRESS' column contains a card for 'Manage product catalog' with a sub-task 'Vendor Storefront ...' and a status of 'MVAECM-6'. The card is assigned to 'Rachit Mane' and has a progress bar at 50%. The 'TO DO' column contains two cards: 'Create artisan storefront' (with sub-task 'Vendor Storefront ...') and 'Browse artisan products' (with sub-task 'Product Discovery...'). Both are assigned to 'Rachit Mane'. The 'DONE' column is empty. On the right side, there is a 'Jira work item' panel showing the 'Activity' section with a comment by 'RACHIT MANE' (RM) stating: 'Started implementation of the product catalog functionality. Working on vendor product management and catalog organization.' The comment was added 1 second ago. The bottom of the screen shows a Windows taskbar with various application icons and a system tray with the date and time (21:57, 31-08-2026).

Figure 12 - Sprint 2 in progress: MVAECM-6 moved to IN PROGRESS. Comment added by RACHIT MANE: "Started implementation of the product catalog functionality. Working on vendor product management and catalog organization."

The screenshot displays the Jira interface for the 'MVAECM board'. The board is organized into four columns: 'TO DO', 'IN PROGRESS', 'IN REVIEW', and 'DONE'. The 'IN PROGRESS' column contains two items: 'Manage product catalog' (MVAECM-6) and 'Create artisan storefront' (MVAECM-5). A comment by RACHIT MANE is visible on the right side of the board.

Board Columns and Items:

- TO DO (2 items):**
 - Browse artisan products (MVAECM-8)
 - View product details (MVAECM-9)
- IN PROGRESS (2 items):**
 - Manage product catalog (MVAECM-6)
 - Create artisan storefront (MVAECM-5)
- IN REVIEW (0 items)**
- DONE (0 items)**

Comment by RACHIT MANE:

Storefront development is in progress. Initial vendor storefront structure and required functionality are being implemented.

Figure 13 - Sprint 2 in progress: MVAECM-5 also moved to IN PROGRESS. Comment added by RACHIT MANE: "Storefront development is in progress. Initial vendor storefront structure and required functionality are being implemented."

← ↻ 🔒 https://rachitmane.atlassian.net/jira/software/c/projects/MVAECM/boards/36?sprintStarted=true

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MVAECM board ...

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Search board RM Version Epic Quick filters Complete sprint Group: Story

TO DO + Create

IN PROGRESS

IN REVIEW

DONE 4

Manage product catalog

- Vendor Storefront & Product Catalog
- MVAECM-6 5

Create artisan storefront

- Vendor Storefront & Product Catalog
- MVAECM-5 3

Browse artisan products

- Product Discovery & Shopping
- MVAECM-8 3

View product details

- Product Discovery & Shopping
- MVAECM-9 3

Type here to search 21:59 31-08-2026

Figure 14 - End of Sprint 2: all four sprint stories (MVAECM-6, MVAECM-5, MVAECM-8 and MVAECM-9) completed in the DONE column.

6. Sprint Summary

The Jira Sprint Report for MVAECM Sprint 1 shows the sprint burndown together with the status report of completed issues. Sprint 1 was completed with all 23 committed story points delivered and no incomplete issues.

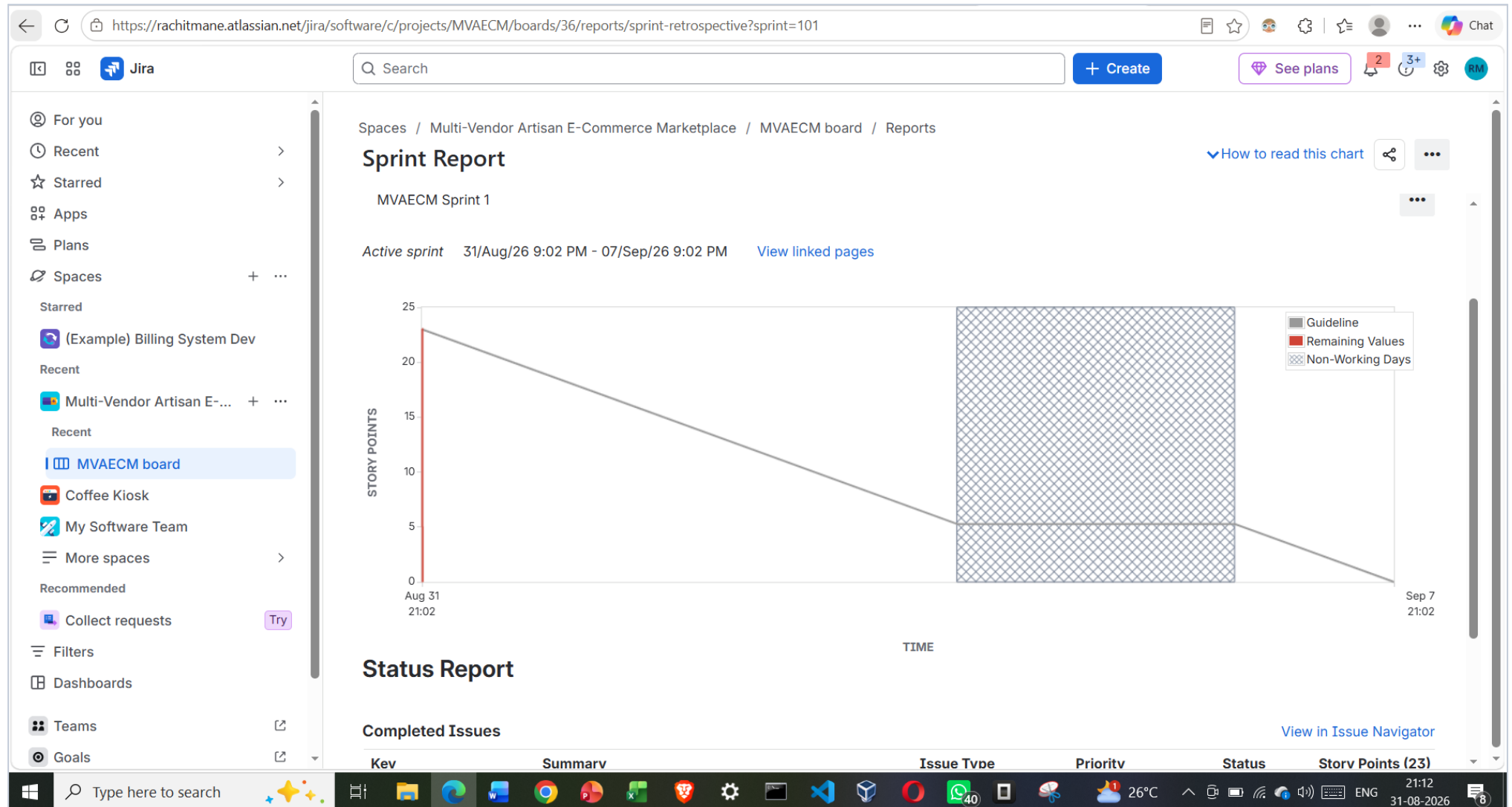


Figure 15 - Sprint Report for MVAECM Sprint 1 while the sprint was active: 31/Aug/26 9:02 PM to 07/Sep/26 9:02 PM.

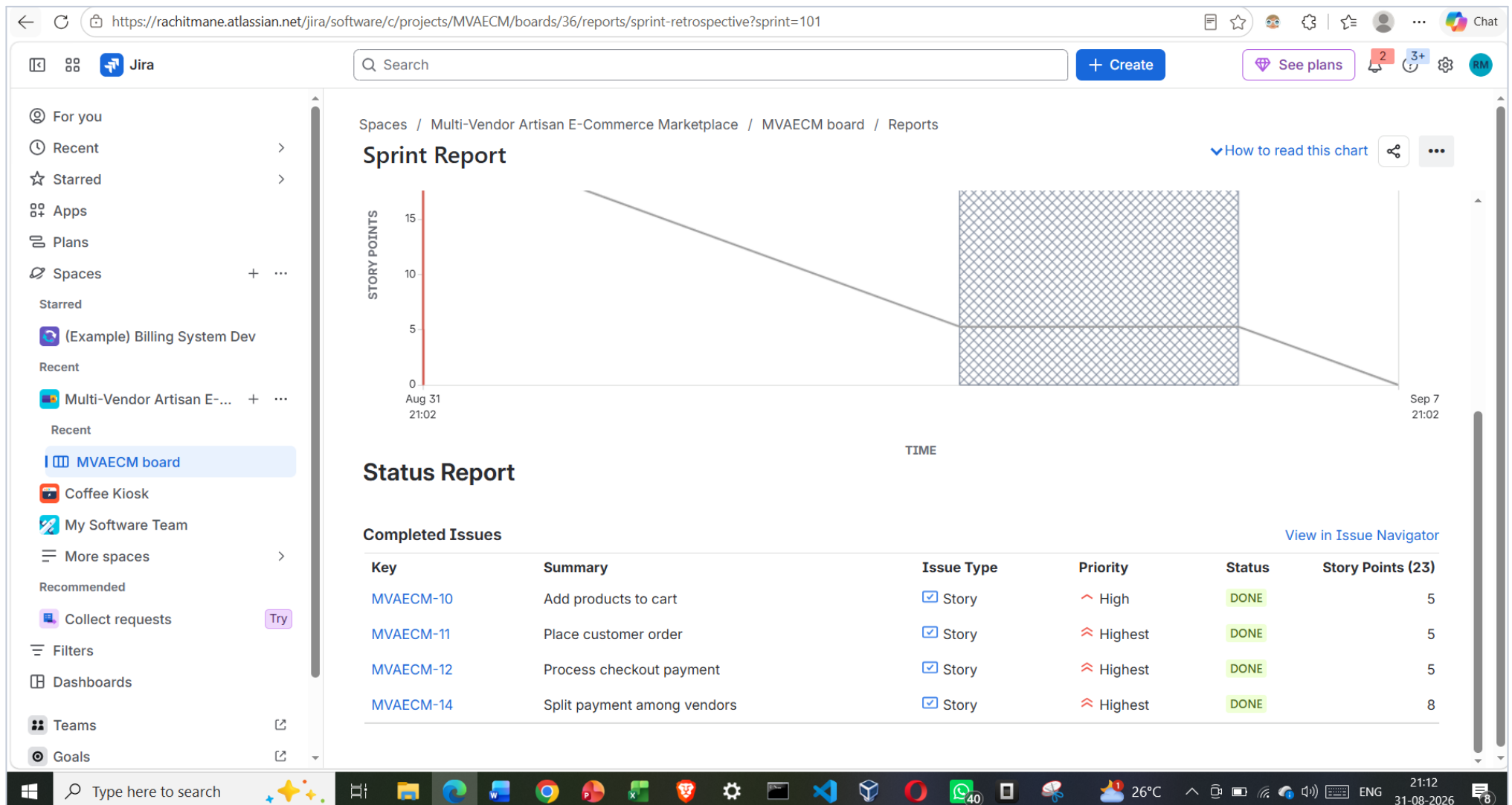


Figure 16 - Sprint Report status report - completed issues: MVAECM-10 Add products to cart (High, 5), MVAECM-11 Place customer order (Highest, 5), MVAECM-12 Process checkout payment (Highest, 5) and MVAECM-14 Split payment among vendors (Highest, 8). Total 23 story points.

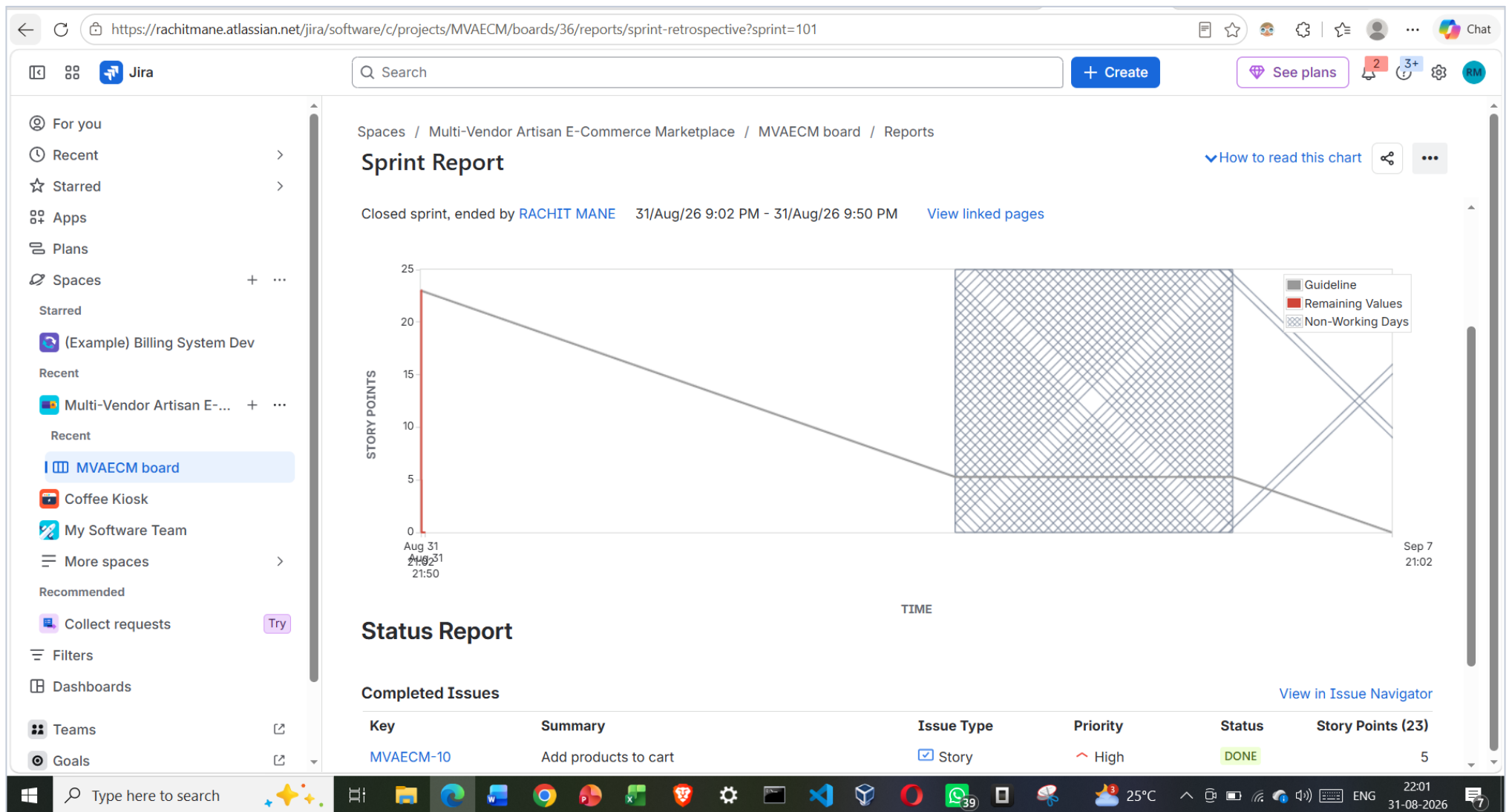


Figure 17 - Sprint 1 completed: "Closed sprint, ended by RACHIT MANE - 31/Aug/26 9:02 PM - 31/Aug/26 9:50 PM", with the completed sprint burndown.

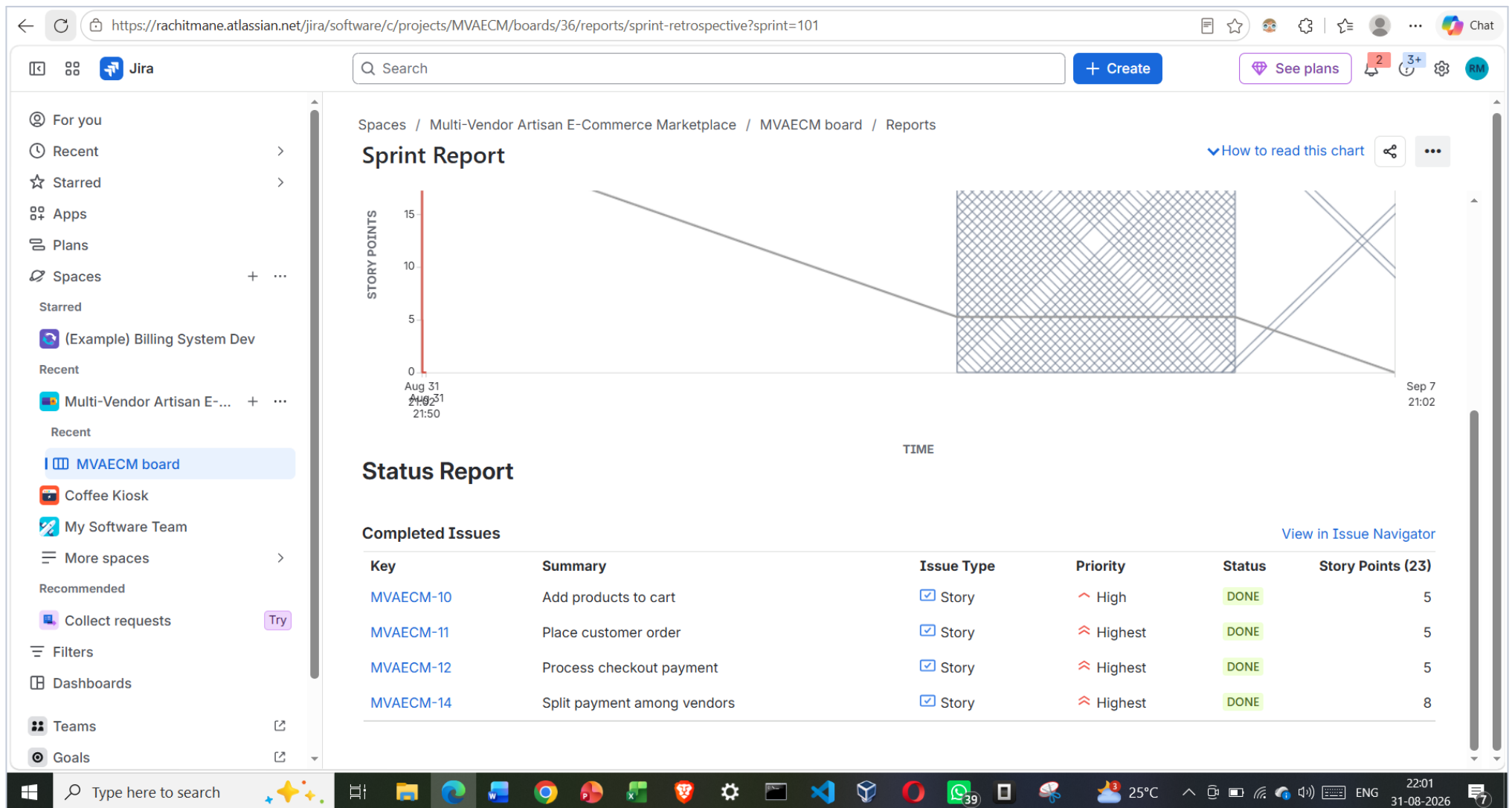


Figure 18 - Sprint 1 completion summary: all four committed stories are listed under Completed Issues with status DONE and a total of 23 story points delivered.

7. Burndown Chart

The burndown chart for each sprint was generated from Jira Reports and plots the remaining story points against time, with the ideal guideline, the remaining values and the non-working days shown.

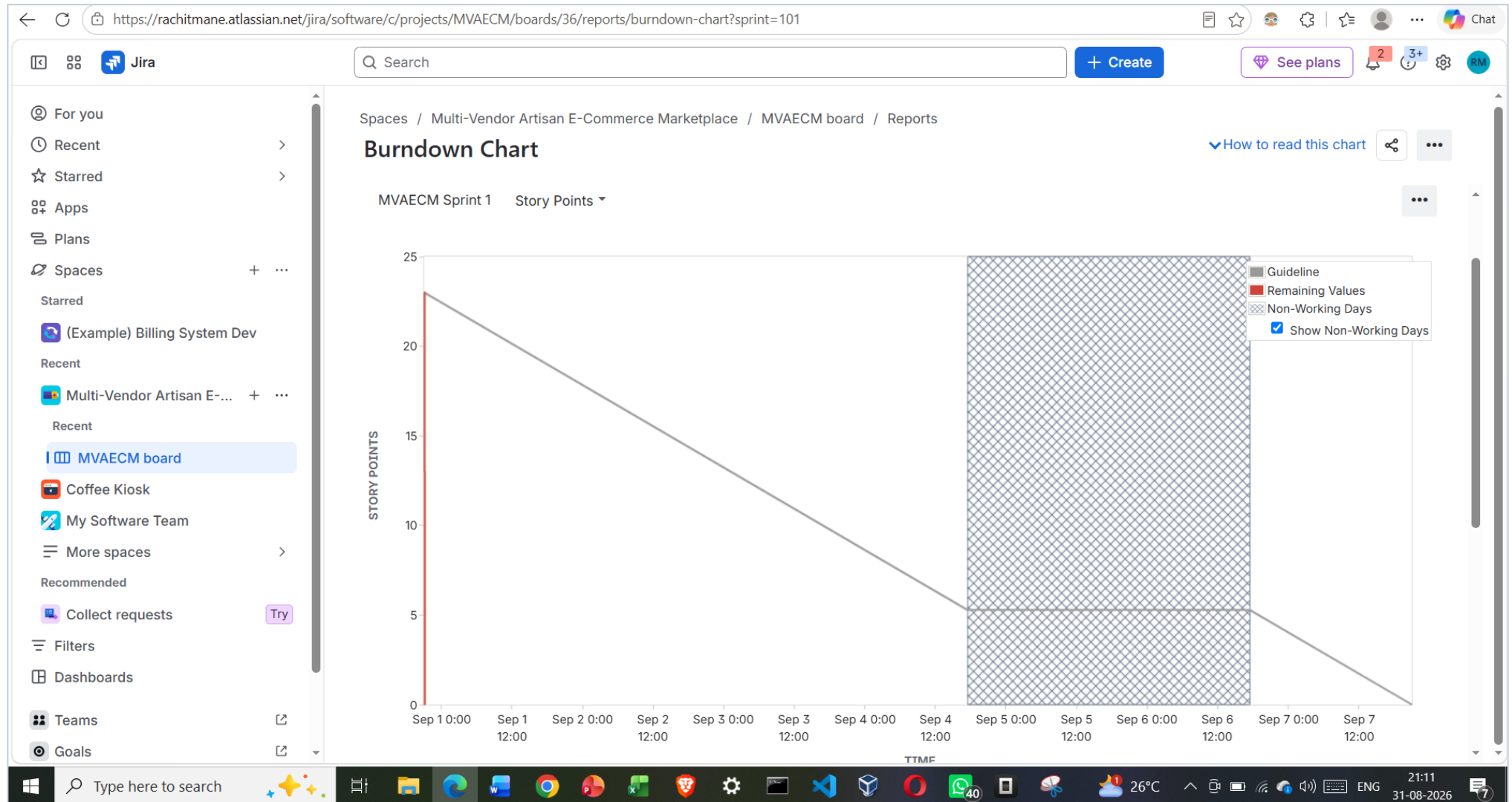


Figure 19 - Burndown chart for MVAECM Sprint 1, measured in story points, starting from the 23 story points committed to the sprint.

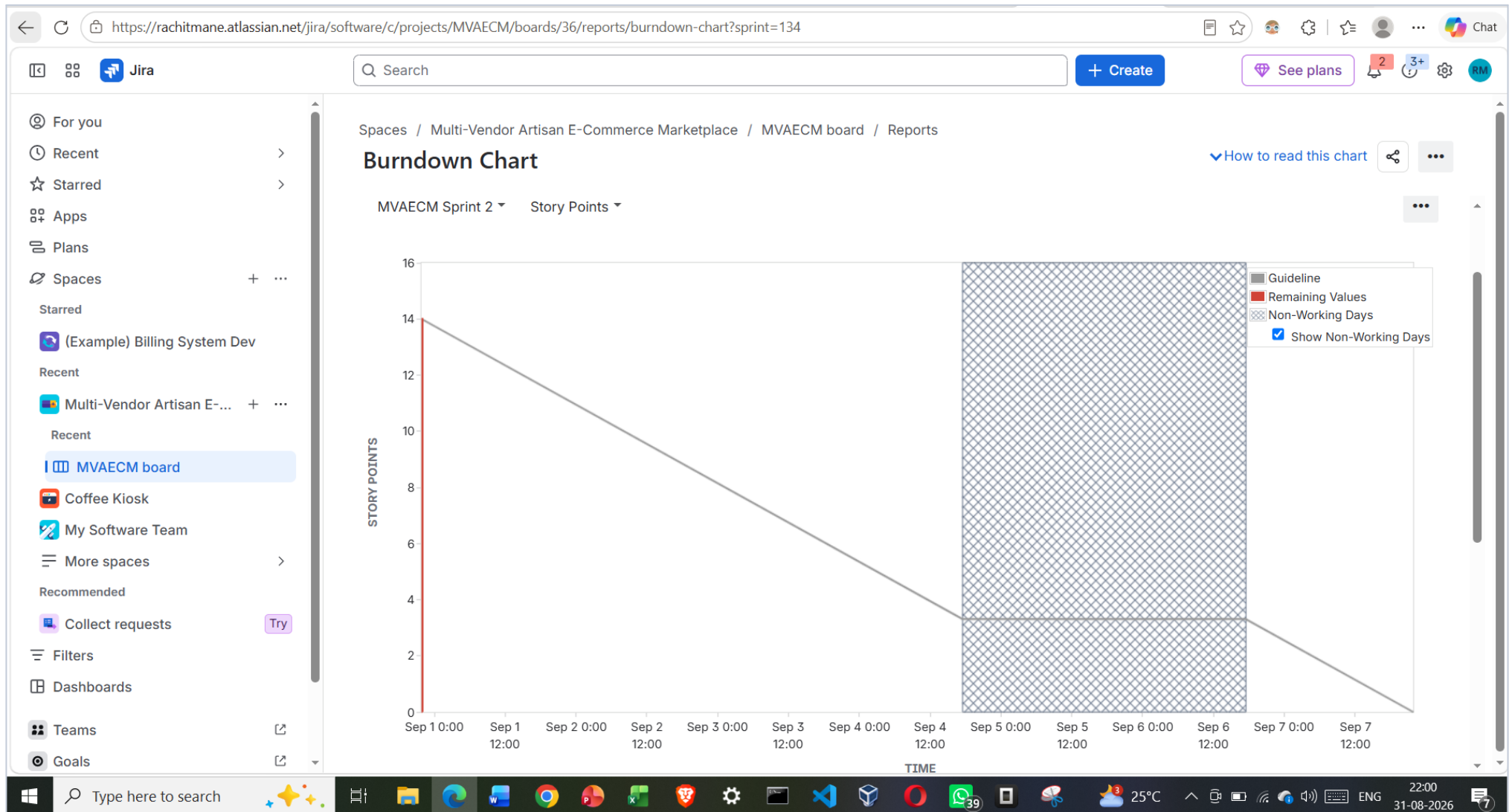


Figure 20 - Burndown chart for MVAECM Sprint 2, measured in story points, starting from the 14 story points committed to the sprint.

8. Group Reflection

To be completed before submission.

No group reflection write-up was present in the repository at the time this document was generated, and no reflection content has been invented here. Please add the group's own reflection text to this section - for example, what went well during the sprint, what did not go well, what the group learned from Planning Poker estimation and backlog ranking, and what the group would do differently in the next sprint.